ref input, SimPoint estimates — small-2way

L1D 32 sets x 2, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	1/1	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	1/1	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	1/1	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	1/1	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	1/1	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	1/1	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	1/1	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	1/1	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	1/1	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	1/1	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	1/1	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	1/1	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	1/1	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	1/1	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	1/1	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	1/1	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	1/1	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	1/1	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	1/1	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	1/1	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	1/1	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	1/1	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	1/1	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	1/1	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	1/1	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	1/1	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	1/1	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	1/1	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	1/1	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	1/1	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	1/1	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	1/1	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	1/1	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	1/1	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.64%	4.39%	33.97%	3.60%	14.03%	64.77%	12.85%	11.23%
parest	done	11.55%	3.60%	12.23%	3.31%	5.40%	63.65%	2.05%	13.78%
deepsjeng	done	9.86%	2.97%	12.82%	7.98%	4.55%	5.80%	1.57%	13.30%
leela	done	6.48%	4.29%	7.20%	0.04%	4.50%	18.20%	0.00%	0.77%
namd	done	12.51%	3.31%	15.86%	1.36%	2.70%	25.47%	5.50%	14.95%
lbm	done	19.64%	0.44%	7.67%	1.75%	81.24%	86.89%	0.00%	0.80%
nab	done	7.99%	1.57%	10.82%	2.82%	0.59%	40.00%	0.77%	0.40%
xz_cld	done	11.84%	3.97%	16.53%	1.39%	4.08%	31.57%	0.01%	0.28%
xz_cpu2006docs	done	9.62%	2.83%	10.47%	2.10%	7.10%	38.70%	3.73%	6.05%
xz_combined	done	10.79%	3.98%	13.39%	1.10%	5.56%	22.05%	0.00%	1.53%
imagick	done	2.17%	3.94%	2.76%	0.03%	0.13%	96.75%	57.27%	20.17%
blender	done	8.22%	2.76%	8.95%	2.06%	3.91%	17.84%	1.16%	32.07%
povray	done	10.32%	7.39%	12.74%	19.42%	4.20%	40.97%	50.50%	60.19%
omnetpp	done	16.65%	0.09%	22.25%	2.02%	5.74%	27.55%	45.06%	1.97%
xalancbmk	done	17.45%	10.52%	20.09%	15.82%	4.71%	68.78%	6.85%	7.90%
x264_pass1	done	2.78%	0.23%	3.36%	0.93%	3.47%	25.29%	3.67%	16.74%
x264_pass2	done	2.28%	0.36%	2.75%	0.87%	2.44%	31.33%	0.45%	16.85%
x264_seek	done	2.30%	0.32%	2.81%	0.97%	2.31%	27.65%	0.54%	12.70%
gcc_pp_O3	done	9.96%	2.05%	13.24%	5.06%	4.18%	37.45%	3.68%	11.87%
gcc_pp_O2	done	10.05%	2.16%	13.34%	5.08%	4.44%	34.58%	3.71%	10.26%
gcc_smaller	done	11.47%	2.78%	14.61%	4.43%	4.02%	43.86%	2.65%	8.57%
gcc_ref32_O5	done	10.16%	3.21%	13.28%	5.14%	4.40%	44.91%	3.95%	10.21%
gcc_ref32_O3	done	11.32%	3.92%	13.63%	6.15%	9.81%	68.76%	2.73%	15.62%
mcf	done	22.05%	7.93%	28.09%	7.61%	4.73%	72.44%	2.72%	16.32%
perlbench_checkspam	done	4.54%	0.85%	7.04%	1.78%	1.66%	16.93%	0.31%	2.13%
perlbench_diffmail	done	6.04%	2.43%	7.82%	7.79%	2.97%	30.54%	0.04%	6.97%
perlbench_splitmail	done	9.96%	0.40%	14.40%	4.28%	3.49%	7.65%	0.22%	0.74%
bwaves_1	done	43.27%	3.94%	47.38%	0.20%	36.83%	24.07%	4.55%	0.30%
bwaves_2	done	42.95%	3.80%	47.10%	0.23%	37.20%	24.39%	4.24%	0.35%
bwaves_3	done	42.88%	4.74%	46.93%	0.17%	36.28%	24.27%	4.24%	0.40%
bwaves_4	done	42.81%	3.89%	46.86%	0.16%	37.10%	24.15%	3.57%	0.27%
exchange2	done	1.83%	1.62%	2.23%	2.78%	1.19%	0.85%	0.08%	0.32%
fotonik3d	done	16.67%	6.14%	22.90%	0.17%	1.24%	41.15%	1.27%	1.81%
roms	done	21.55%	4.01%	26.62%	1.25%	15.16%	38.82%	7.76%	5.84%
wrf	done	17.88%	9.00%	18.88%	5.95%	15.69%	21.73%	6.33%	8.69%
cam4	done	11.80%	3.18%	12.89%	3.08%	12.83%	50.53%	5.20%	19.51%
pooled		12.40%	3.33%	15.42%	2.81%	7.56%	41.46%	8.56%	11.14%
mean		14.31%	3.42%	17.05%	3.58%	10.83%	37.23%	6.92%	9.77%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.